Second-Order Methods

Newton, Gauss-Newton, and Levenberg–Marquardt

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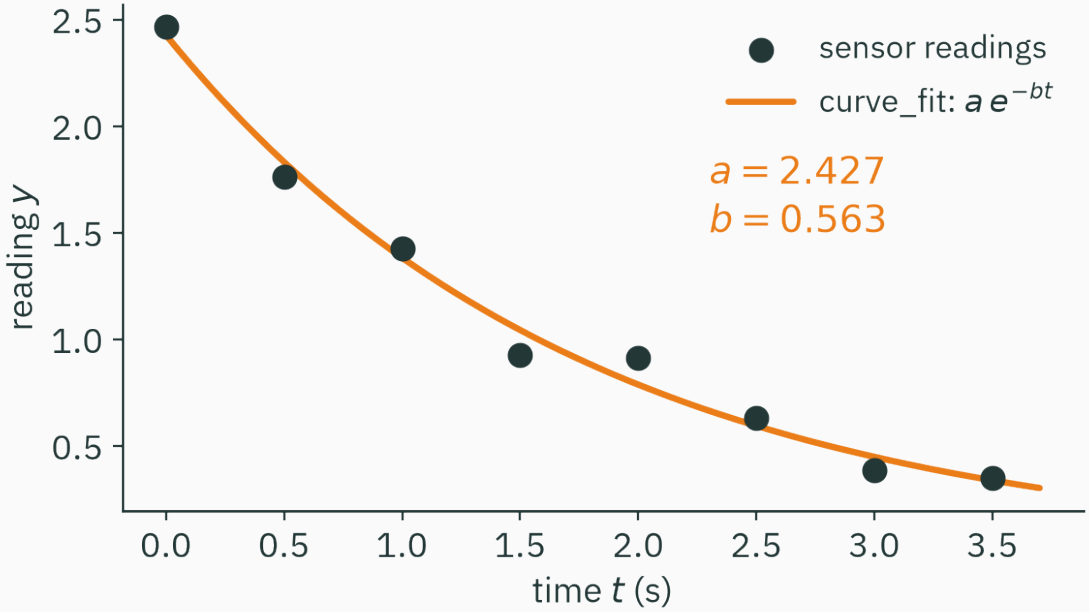
Mathematical Foundations for AI · IIT Gandhinagar

Nonlinear least squares in `scipy`

A sensor, a decay, eight points

V VISUAL

You're calibrating a sensor. After a pulse, its reading decays like $a\,e^{-bt}$ — and the datasheet needs the decay rate b. You log eight noisy readings:



Two unknowns (a, b), eight equations that don't quite agree. Every lab on earth solves this the same way...

Everyone calls the same one-liner

```
from scipy.optimize import curve_fit

def model(t, a, b):
    return a * np.exp(-b * t)

popt, pcov = curve_fit(model, t, y, p0=(1.0, 0.2))
# popt = [2.4274, 0.5628]
```

No learning rate. No 10,000 iterations. It lands on the answer in **about six steps**.

`method='lm'` selects Levenberg–Marquardt. We will derive the method and reproduce these fitted parameters.

From a linear model to a quadratic model

L17′s gradient descent trusted a **line** — and paid for it in steps, with the condition number κ setting the price.

A quadratic local model has a finite minimizer; solving for it gives the Newton step.

- **Newton**: model the loss with L9′s quadratic Taylor, jump to the model's minimum.
- **Gauss-Newton**: when the loss is a **sum of squares**, get the curvature almost free.
- **Levenberg–Marquardt**: damp the Gauss-Newton system when its local model is unreliable.

Learning outcomes

By the end of this lecture you will be able to:

1. Derive ⭐ the Newton step $\boldsymbol{\delta} = -H^{-1}\nabla f$ by minimizing the quadratic model.
2. Explain why Newton finishes **any** quadratic in one step — for **any** condition number.
3. Run Newton by hand in 1-D and recognize **quadratic convergence** (digits double).
4. Price a Newton step (d^2 memory, d^3 solve) and name its failure modes (saddles, ascent).
5. Set up **nonlinear least squares**: residual vector $\boldsymbol{r}$, its Jacobian J, $\nabla\mathcal{L} = J^\top \boldsymbol{r}$.
6. Derive the **Gauss-Newton** system $(J^\top J)\boldsymbol{\delta} = -J^\top \boldsymbol{r}$ and interpret LM's damping parameter λ.
7. Say precisely what `scipy.optimize.curve_fit` runs.

A parabola instead of a line

L17 in one slide: the line and its price

Gradient descent came from trusting the **first-order** Taylor model for one small step:

$$f(\boldsymbol{x}+\boldsymbol{\delta}) \approx f(\boldsymbol{x}) + \nabla f(\boldsymbol{x})^\top \boldsymbol{\delta} \quad \Rightarrow \quad \boldsymbol{\delta} = -\eta \nabla f(\boldsymbol{x})$$

- The learning rate η = how far you trust a line that is only true at a point.
- On a bowl with condition number κ: the safe η is set by the **steepest** wall, so the **flattest** direction crawls — the step budget grows like κ. That was the main result from L17.

A line has no bottom. However far you trust it, it only ever says “downhill is that way” — never “stop **here**”.

The second-order Taylor model has a minimizer

L9 built the **second-order** Taylor model and promised it would run Module 4:

$$f(\boldsymbol{x}+\boldsymbol{\delta}) \approx \underbrace{f(\boldsymbol{x})}_{\text{height}} + \underbrace{\nabla f(\boldsymbol{x})^{\top}\boldsymbol{\delta}}_{\text{tilt}} + \underbrace{\frac{1}{2}\boldsymbol{\delta}^{\top} H(\boldsymbol{x})\,\boldsymbol{\delta}}_{\text{bend}}$$

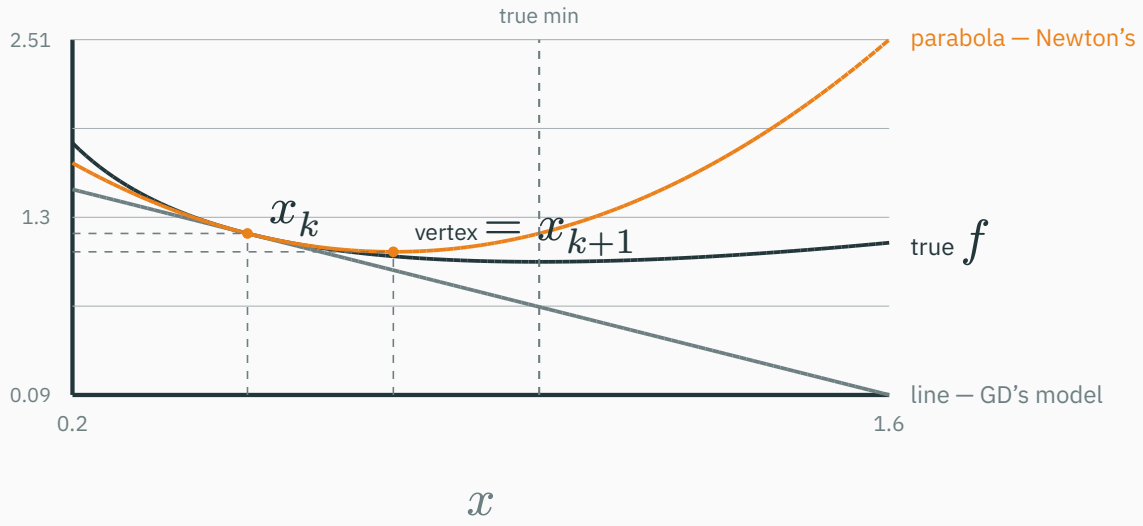
- GD kept height + tilt. Today we keep the **bend** too.
- If H is positive definite (L9: all eigenvalues > 0), this model is a **bowl** — and a bowl has a **bottom you can name**.

New policy: don't step along the model — jump to the model's minimum.

Picture first: two models, two policies

V VISUAL

Same function, same point, both local stories drawn at $x_k = 0.5$ (here $f(x) = x - \ln x$):



the line never bottoms out — **you** must choose a step; the parabola has a vertex — **it** chooses the step for you

The two policies, side by side

	gradient descent (L17)	**Newton (today)**
local model	line: $f + \nabla f^\top \boldsymbol{\delta}$	bowl: $f + \nabla f^\top \boldsymbol{\delta} + \frac{1}{2}\boldsymbol{\delta}^\top H \boldsymbol{\delta}$
the step	$-\eta \nabla f$ — direction from the model, **length from you**	jump to the model's minimum — **no η anywhere**
uses	slopes only	slopes **and** curvature

One question remains: **where exactly is the bowl's bottom?** That formula is today's ⭐.

The Newton step

⭐ Step 1 — in 1-D: minimize the parabola

D DERIVATION

The model around x_k, as a function of the step δ:

$$q(\delta) = f(x_k) + f'(x_k)\,\delta + \frac{1}{2} f''(x_k)\,\delta^2$$

A parabola in δ — minimize it like any 1-D function (set $\frac{\mathrm{d}q}{\mathrm{d}\delta} = 0$, L7):

$$f'(x_k) + f''(x_k)\,\delta = 0 \quad \Rightarrow \quad \delta^* = -\frac{f'(x_k)}{f''(x_k)}$$

Numbers — $f(x) = x - \ln x$ at $x_k = 0.5$: $f' = 1 - \frac{1}{0.5} = -1$, $f'' = \frac{1}{0.5^2} = 4$:

$$\delta^* = -\frac{-1}{4} = 0.25 \quad \Rightarrow \quad x_{k+1} = 0.75 \quad \text{— the vertex in the picture, exactly}$$

⭐ Step 2 — in n-D: same move, matrix clothes

D DERIVATION

$$q(\boldsymbol{\delta}) = f(\boldsymbol{x}_k) + \nabla f^\top \boldsymbol{\delta} + \frac{1}{2}\boldsymbol{\delta}^\top H \boldsymbol{\delta}$$

Differentiate with respect to $\boldsymbol{\delta}$ using L9′s two gradient rules ($\nabla(\boldsymbol{g}^\top \boldsymbol{\delta}) = \boldsymbol{g}$, and the ⭐ rule $\nabla\left(\frac{1}{2}\boldsymbol{\delta}^\top H \boldsymbol{\delta}\right) = H\boldsymbol{\delta}$ for symmetric H):

$$\nabla_{\boldsymbol{\delta}}\, q = \nabla f + H\boldsymbol{\delta} = \mathbf{0}$$

The Newton system: $H\boldsymbol{\delta} = -\nabla f$**, so** $\boldsymbol{\delta}_{\text{Newton}} = -H^{-1}\nabla f$

Fine print: this stationary point is the model's **minimum** only when H is positive definite — L9′s bowl test. Pin that; it becomes a plot twist in Section 5.

Read the step like an engineer

$$\underbrace{\boldsymbol{\delta} = -\eta\,\nabla f}_{\text{GD}} \qquad \text{vs} \qquad \underbrace{\boldsymbol{\delta} = -H^{-1}\,\nabla f}_{\text{Newton}}$$

- Newton is GD with the scalar η promoted to a **matrix-valued learning rate** H^{-1}.
- In the eigenbasis (L5): direction i gets step $1/\lambda_i$ — steep walls small steps, flat valleys big ones. Exactly what L17 begged for, **per direction, automatically**.
- In practice you never form H^{-1}: **solve** $H\boldsymbol{\delta} = -\nabla f$ (Module 1 discipline — solving is cheaper and more stable than inverting).

Numbers: one Newton step on L9's bowl

$f(x,y) = x^2 + xy + y^2$ at $(1,1)$ — the running function of L9, with its old friends:

$$\nabla f = \begin{pmatrix} 2x + y \\ x + 2y \end{pmatrix} = \begin{pmatrix} 3 \\ 3 \end{pmatrix}, \qquad H = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \text{ (everywhere)}$$

Solve $H\boldsymbol{\delta} = -\nabla f$ by hand (2×2 elimination):

$$\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \boldsymbol{\delta} = \begin{pmatrix} -3 \\ -3 \end{pmatrix} \quad \Rightarrow \quad \boldsymbol{\delta} = \begin{pmatrix} -1 \\ -1 \end{pmatrix} \quad \Rightarrow \quad \boldsymbol{x}_1 = (0,0)$$

And $\nabla f(0,0) = \mathbf{0}$ — the **exact minimum, in one step**. (Chalkdust's solver agrees, live: $\boldsymbol{\delta} = \begin{pmatrix} -1 \\ -1 \end{pmatrix}$.) Luck? Next slide: it can't miss.

On any quadratic, Newton cannot miss

D DERIVATION

Take **any** quadratic $f(\boldsymbol{x}) = \frac{1}{2}\boldsymbol{x}^\top A\boldsymbol{x} + \boldsymbol{b}^\top \boldsymbol{x} + c$ with A positive definite. Then (L9 rules):

$$\nabla f = A\boldsymbol{x} + \boldsymbol{b}, \qquad H = A \quad \text{— the same matrix at every point}$$

So from **any** starting point $\boldsymbol{x}_0$:

$$\boldsymbol{x}_1 = \boldsymbol{x}_0 - A^{-1}(A\boldsymbol{x}_0 + \boldsymbol{b}) = \boldsymbol{x}_0 - \boldsymbol{x}_0 - A^{-1}\boldsymbol{b} = -A^{-1}\boldsymbol{b}$$

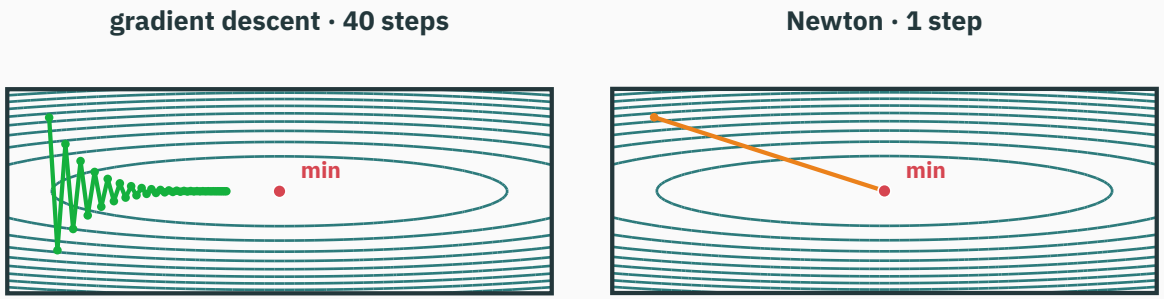
— which is exactly where $\nabla f = 0$: **the** minimum.

On a quadratic, the model is the function — one jump, done, and κ never entered the formula.

Compare gradient descent and Newton on $\kappa = 50$

V VISUAL

Use the same start on $f = \frac{1}{2}(x^2 + 50y^2)$; both trajectories are computed in the slide:



GD: zigzags die out, then a long crawl — 40 steps and still $\sim 23\%$ of the x-gap left. Newton: $H\delta = -\nabla f$, solved once.

Crank κ from 50 to 5000: GD's step budget grows $\sim 100\times$ (L17). Newton's budget: still **one**. Its step is built from H, so conditioning is priced in.

Code: Newton in five lines

```
def newton(grad, hess, x, steps=10):
    for _ in range(steps):
        x = x - np.linalg.solve(hess(x), grad(x))   # solve, never invert
    return x

A = np.array([[1., 0.], [0., 50.]])                  # the kappa = 50 bowl
newton(lambda x: A @ x, lambda x: A, np.array([-2.2, 0.65]), steps=1)
# array([0., 0.])          <- one step, exact
```

Compare L17's tuning ritual: no `lr=`, no schedule, no divergence watch. The price appears later — first, let's enjoy it on functions that **aren't** parabolas.

Checkpoint: Newton on a quadratic

Q QUESTION

Newton's method is applied to $f(x) = 3(x-2)^2$ starting at $x_0 = 10$. Where is x_1?

A. $x_1 = 2$

B. $x_1 = 6$ — halfway

C. $x_1 = 4$

D. Cannot say — depends on the learning rate

Answer: one step reaches the quadratic minimum

A ANSWER

Correct: A — $x_1 = 2$, the exact minimum

Why: f is quadratic, so the second-order model equals f: $f'(10) = 6(10 - 2) = 48$, $f'' = 6$, $\delta = -\frac{48}{6} = -8$, and $x_1 = 2$. Newton has no learning rate; the curvature f'' scales the gradient.

Newton beyond quadratics

Not a parabola? Re-model and jump again

For a non-quadratic objective, the model is local, so recompute it after every step:

$$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k - H(\boldsymbol{x}_k)^{-1}\nabla f(\boldsymbol{x}_k) \qquad (\text{1-D: } x_{k+1} = x_k - f'(x_k)/f''(x_k))$$

Each jump lands at the **model's** bottom, not the function's — then fits a fresh parabola there.

Our 1-D running example $f(x) = x - \ln x$ (min at $x^* = 1$): the update simplifies beautifully —

$$x_{k+1} = x_k - \frac{1 - 1/x_k}{1/x_k^2} = x_k - (x_k^2 - x_k) = 2x_k - x_k^2$$

No calculus left at run time. Let's run it.

You've met Newton before

Newton's method was born for **root finding** — solve $g(x) = 0$ by following the tangent:

$$x_{k+1} = x_k - \frac{g(x_k)}{g'(x_k)} \qquad \text{(Newton–Raphson, the school version: } \sqrt{2} \text{ via } x_{k+1} = (x_k + 2/x_k)/2 \text{)}$$

Minimizing f means solving $\nabla f = \mathbf{0}$. Substitute $g = f'$:

$$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)} \quad \text{— the same formula we just derived from the parabola}$$

Two readings, one method: **jump to the parabola's vertex** $\equiv$ **chase a zero of the slope**. The second reading explains the dangers coming in Section 5: a zero of ∇f need not be a minimum...

Watch the digits double

$x_{k+1} = 2x_k - x_k^2$ from $x_0 = 0.5$, run live in the slide (target $x^* = 1$):

k	x_k	error $\lvert x_k - 1 \rvert$	correct digits ($-\log_{10}$)
0	0.5000000000	5.0×10^{-1}	0.3
1	0.7500000000	2.5×10^{-1}	0.6
2	0.9375000000	6.3×10^{-2}	1.2
3	0.9960937500	3.9×10^{-3}	2.4
4	0.9999847412	1.5×10^{-5}	4.8
5	0.9999999998	2.3×10^{-10}	9.6

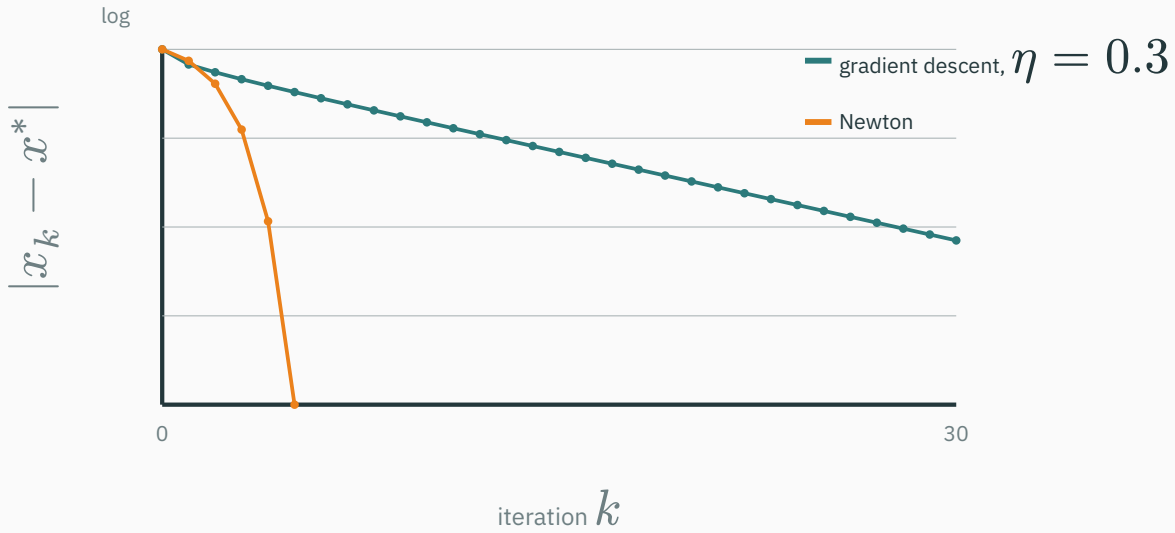
The correct-digit count $0.3 \to 0.6 \to 1.2 \to 2.4 \to 4.8 \to 9.6$ — **doubling every step**.

Here it's exact: $x_{k+1} - 1 = 2x_k - x_k^2 - 1 = -(x_k - 1)^2$, so $e_{k+1} = e_k^2$. The error squares.

The race on log paper

V VISUAL

Same function, same start: GD ($\eta = 0.3$) vs Newton, error per iteration — log scale:



GD is a **straight line** on log paper — a constant factor (0.7) per step. Newton's slope itself steepens: that's the error squaring. 10 digits by $k = 5$ vs 5 digits by $k = 30$.

Quadratic convergence, said honestly

Near a minimum with H positive definite and smooth curvature:

$$\|e_{k+1}\| \le C\|e_k\|^2 \qquad \text{— "quadratic convergence": digits double per step}$$

- GD's **linear** convergence multiplies the error by a constant (L17: $1 - \frac{1}{\kappa}$, painful for large κ); Newton **squares** it — and squaring doesn't care about κ.
- The qualification is **near**: far from the minimum, the quadratic approximation may be inaccurate.

Newton converges rapidly when initialized near a suitable minimum. Far away, the step can increase the objective or converge to another stationary point.

Computational cost and failure modes

Computational cost of one Newton step

For d parameters, per iteration:

	needs	memory	arithmetic
gradient descent	∇f	$O(d)$	$O(d)$ once ∇f is in hand
Newton	∇f, H, a solve	$O(d^2)$	$O(d^3)$ for the solve

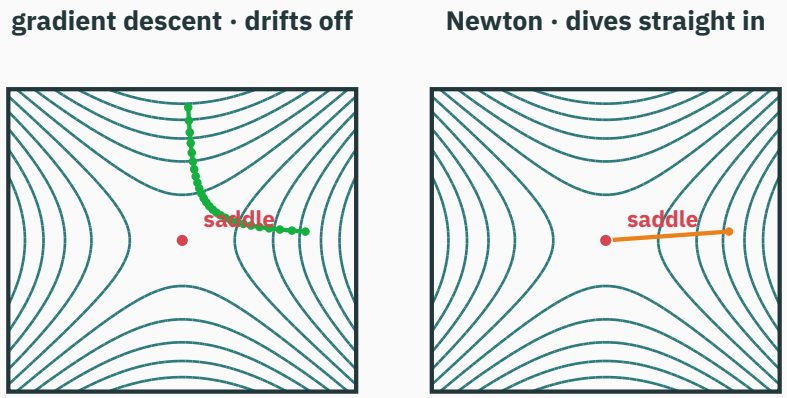
At L9′s modest network, $d = 10^6$: H has 10^{12} entries $\approx$ **4 TB** in float32 (L9′s warning, now due) — and one $O(d^3) = 10^{18}$-FLOP solve is **hours on a modern GPU, for a single step**.

Newton's genius is per-step; its cost is per-dimension, cubed. In 2-D it's free. At 10^6-D it's fantasy.

Newton can converge to a saddle point

V VISUAL

Newton chases $\nabla f = \mathbf{0}$ — and (L16) a saddle **is** a $\nabla f = \mathbf{0}$ point. On $f = \frac{1}{2}(x^2 - y^2)$:

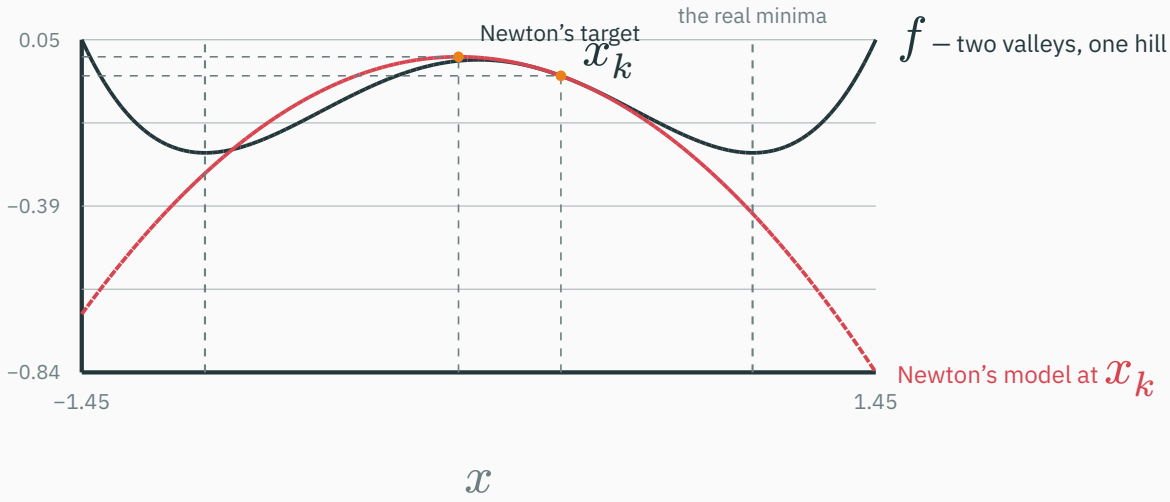


GD leaks along $-y$ and escapes downhill; Newton's model of a saddle is the saddle — its "jump to the stationary point" lands **on** it, in one step

An indefinite Hessian can produce an ascent direction

V VISUAL

On a double well, stand at $x_k = 0.3$ where $f'' = -0.73 < 0$ — the parabola **frowns**:



a frowning parabola's stationary point is its **peak**: the "Newton step" here moves **left, toward the hilltop at** $x = 0$ — and iterating converges to that local maximum

When H is not positive definite, $-H^{-1}\nabla f$ need not point downhill. Newton without a safety check is only trustworthy where the world is bowl-shaped.

Damping, line search, and trust regions

All practical second-order methods are Newton plus a seatbelt:

- **Damping**: solve $(H + \tau I)\,\delta = -\nabla f$ — adding τI lifts every eigenvalue by τ, forcing the model convex (L5: eigenvalues shift together).
- **Line search**: take the Newton **direction**, but test the length before committing.
- **Trust region**: only believe the model inside a radius; grow or shrink the radius by results.

Hold the "$+\tau I$" thought. In twenty minutes it returns wearing a famous name — and it's the exact knob inside `curve_fit`.

Sums of squares: nonlinear least squares

Back to the sensor: the loss is a sum of squares

Our eight readings, and the model $\hat{y}(t) = a\,e^{-bt}$ with $\boldsymbol{\theta} = (a, b)$:

t_i	0.0	0.5	1.0	1.5	2.0	2.5	3.0	3.5
y_i	2.465	1.762	1.426	0.928	0.914	0.632	0.385	0.352

Per point, a **residual** — the model's miss; the loss adds the squares (L14: this is the Gaussian NLL):

$$r_i(\boldsymbol{\theta}) = a\,e^{-bt_i} - y_i, \qquad \mathcal{L}(\boldsymbol{\theta}) = \frac{1}{2}\sum_{i=1}^{8} r_i(\boldsymbol{\theta})^2 = \frac{1}{2}\|\boldsymbol{r}(\boldsymbol{\theta})\|^2$$

The $\frac{1}{2}$ is cosmetics: it cancels the 2 that differentiation drops. This **shape** — squared residuals — is everywhere: calibration, GPS, camera bundle adjustment, regression.

The residual vector and its Jacobian

$\boldsymbol{r} : \mathbb{R}^2 \to \mathbb{R}^8$ — two parameters in, eight misses out. Its derivative is a **Jacobian** (L9): one row per residual, one column per parameter, 8×2:

$$J_{i1} = \frac{\partial r_i}{\partial a} = e^{-bt_i}, \qquad J_{i2} = \frac{\partial r_i}{\partial b} = -a\, t_i\, e^{-bt_i}$$

At the bad first guess $\boldsymbol{\theta}_0 = (1.0, 0.2)$, three of the eight rows:

row	t_i	$\partial r_i/\partial a$	$\partial r_i/\partial b$
$i = 1$	0.0	1.000	0.000
$i = 2$	0.5	0.905	-0.452
$i = 8$	3.5	0.497	-1.738

Late-time rows barely feel a but feel b strongly — the Jacobian **is** the sensitivity table.

The gradient, in Jacobian clothes

Chain rule on $\mathcal{L} = \frac{1}{2}\boldsymbol{r}^\top \boldsymbol{r}$, shapes doing the bookkeeping (L9′s habit):

$$\nabla \mathcal{L} = \underbrace{J^\top}_{2\times 8}\underbrace{\boldsymbol{r}}_{8\times 1} \quad \in \mathbb{R}^2$$

At $\boldsymbol{\theta}_0 = (1.0, 0.2)$, with the measured residuals: $\nabla \mathcal{L} = J^\top \boldsymbol{r} = (-2.894, 0.937)$.

- Gradient descent is ready to go: $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} - \eta\, J^\top \boldsymbol{r}$. L17 machinery, nothing new.
- But Newton wants H — second derivatives of all eight r_i. Expensive… **unless the squares themselves hand us the curvature.** They do.

Gauss-Newton: curvature almost for free

The idea: linearize the residuals, not the loss

Replace each residual by its L9 linear story at the current $\boldsymbol{\theta}$:

$$\boldsymbol{r}(\boldsymbol{\theta}+\boldsymbol{\delta}) \approx \boldsymbol{r} + J\boldsymbol{\delta} \quad \Rightarrow \quad \mathcal{L}(\boldsymbol{\theta}+\boldsymbol{\delta}) \approx \frac{1}{2}\|\boldsymbol{r} + J\boldsymbol{\delta}\|^2$$

Look at what the right side **is**: a **linear least-squares problem in δ** — the one problem Module 1 taught us to solve exactly (and L16 proved convex).

A line inside a square is a parabola: linearizing r hands us a quadratic model of $\mathcal{L}$ — without ever touching a second derivative.

The Gauss-Newton step

Minimize $\frac{1}{2}\|\boldsymbol{r} + J\boldsymbol{\delta}\|^2$ over $\boldsymbol{\delta}$: its gradient is $J^\top(\boldsymbol{r} + J\boldsymbol{\delta}) = \mathbf{0}$ — the **normal equations** of the linearized fit:

Gauss-Newton: solve $(J^\top J)\,\boldsymbol{\delta} = -J^\top \boldsymbol{r}$**, then** $\boldsymbol{\theta} \leftarrow \boldsymbol{\theta} + \boldsymbol{\delta}$**; repeat.**

Pattern-match against Newton's $H\boldsymbol{\delta} = -\nabla f$:

- right side: $J^\top \boldsymbol{r} = \nabla \mathcal{L}$ ✓ (last slide) — same gradient,
- left side: $J^\top J$ **standing where the Hessian stands**. Is it the Hessian? Almost —

Why this is secretly Newton

Differentiate $\nabla \mathcal{L} = J^\top r$ once more (product rule — each factor depends on θ):

$$H = J^\top J + \sum_{i=1}^{n} r_i \, \nabla^2 r_i$$

Gauss-Newton keeps the first term, **drops the second**. When is that fair?

- $r_i \approx 0$ near a good fit — the dropped term is weighted by the **misses**;
- or r_i nearly linear ($\nabla^2 r_i \approx 0$) — nothing to drop.

$J^\top J$ = curvature from first derivatives only — and it's always PSD, so the model is never a frown. Newton's power, GD's ingredients.

Numbers: the first step from $\boldsymbol{\theta}_0 = (1.0, 0.2)$

Assemble the 2×2 system from the real data (all eight rows summed — check any entry by hand):

$$J^\top J = \begin{pmatrix} 4.403 & -5.488 \\ -5.488 & 11.938 \end{pmatrix}, \qquad J^\top \boldsymbol{r} = \begin{pmatrix} -2.894 \\ 0.937 \end{pmatrix}$$

$$\begin{pmatrix} 4.403 & -5.488 \\ -5.488 & 11.938 \end{pmatrix} \boldsymbol{\delta} = \begin{pmatrix} 2.894 \\ -0.937 \end{pmatrix} \quad \Rightarrow \quad \boldsymbol{\delta} = \begin{pmatrix} 1.310 \\ 0.524 \end{pmatrix} \quad \Rightarrow \quad \boldsymbol{\theta}_1 = (2.310, 0.724)$$

From $(1.0, 0.2)$ straight to $(2.31, 0.72)$ — one solve of a 2×2 system.

---

b overshoots the fitted value 0.563 on the first step because the residual linearization is local. Recompute it and continue.

The run: five solves to machine precision

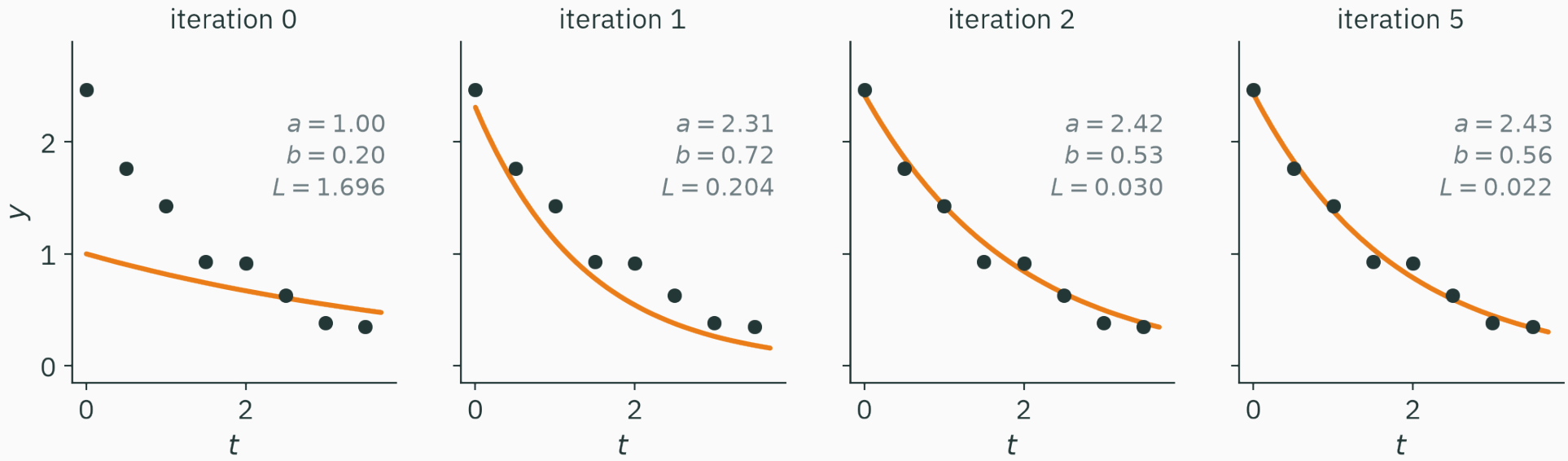
The whole optimization, on the real data (this table is the figure script's actual output):

k	a_k	b_k	**loss $\mathcal{L}(\theta_k)$**
0	1.000000	0.200000	1.69626671
1	2.309903	0.723655	0.20392293
2	2.418643	0.525822	0.03006333
3	2.425973	0.560925	0.02159785
4	2.427402	0.562773	0.02158193
5	**2.427436**	**0.562798**	**0.02158193**

b rings — overshoot, undershoot, settle — as each linearization corrects the last. By $k = 5$ the loss has stopped changing at 8 decimals: the digits-double regime again.

The fit, evolving

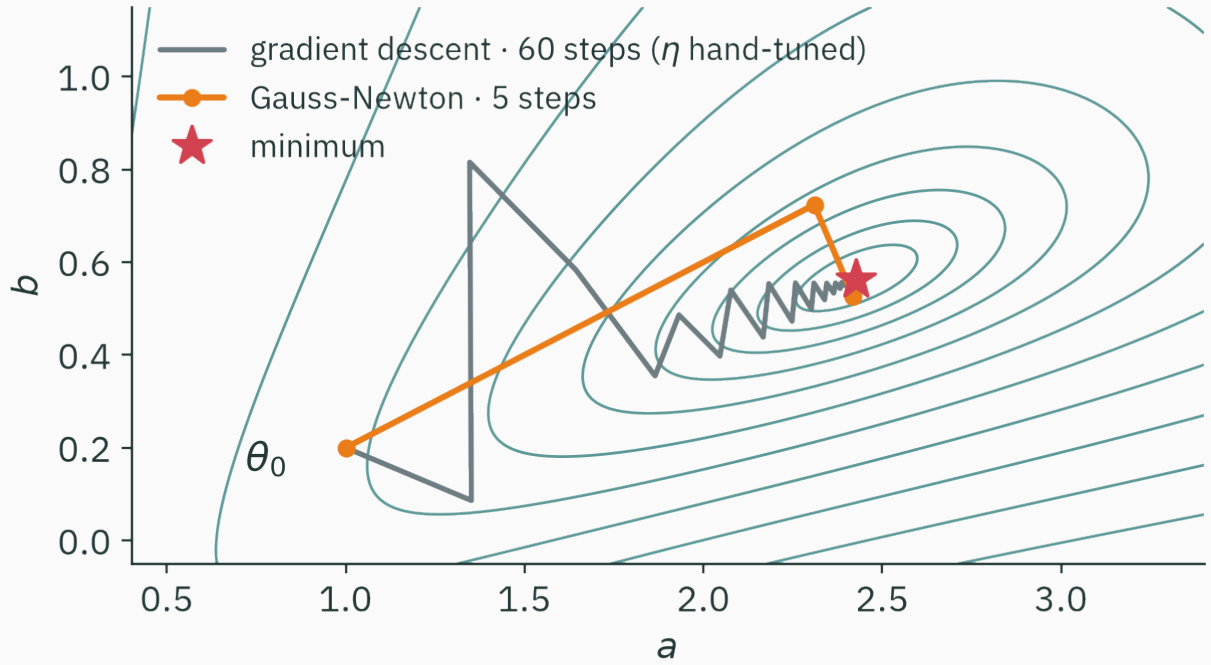
V VISUAL



The first solve moves close to the data; by iteration 2 the remaining error is comparable to the observation noise.

Parameter trajectory in the (a, b) plane

V VISUAL



gradient descent zigzags into the curved valley — and its η was hand-tuned **using $J^\top J$'s eigenvalues**; 25% more η and it never converges. Gauss-Newton: two visible hops, three invisible ones.

Code: Gauss-Newton in eight lines

```
theta = np.array([1.0, 0.2])                  # (a, b) — the bad guess
for _ in range(6):
    a, b = theta
    r = a * np.exp(-b * t) - y                # residuals        (8,)
    J = np.stack([np.exp(-b * t),             # dr/da            (8, 2)
                  -a * t * np.exp(-b * t)], axis=1)
    delta = np.linalg.solve(J.T @ J, -J.T @ r)
    theta = theta + delta
# theta = [2.427437, 0.562798]
```

Residuals, Jacobian, one small solve — no Hessian code anywhere, and no learning rate.

Levenberg–Marquardt damping

When the Gauss-Newton model is unreliable

Gauss-Newton can fail when its local residual model is inaccurate or poorly conditioned:

- Far from the fit, the linearization lies — our very first step overshot b by 29%.
- If the data barely constrains a parameter, $J^\top J$ is **nearly singular** (L6: tiny singular values) — the solve amplifies noise into a **wild** step.
- Nothing stops a step from **increasing** the loss. There is no seatbelt.

Section 5 already prescribed the seatbelt: **damp it** — add λI before solving.

The damping parameter λ

Levenberg–Marquardt: $\left(J^\top J + \lambda I\right)\boldsymbol{\delta} = -J^\top \boldsymbol{r}$

The two limiting cases explain the effect of λ:

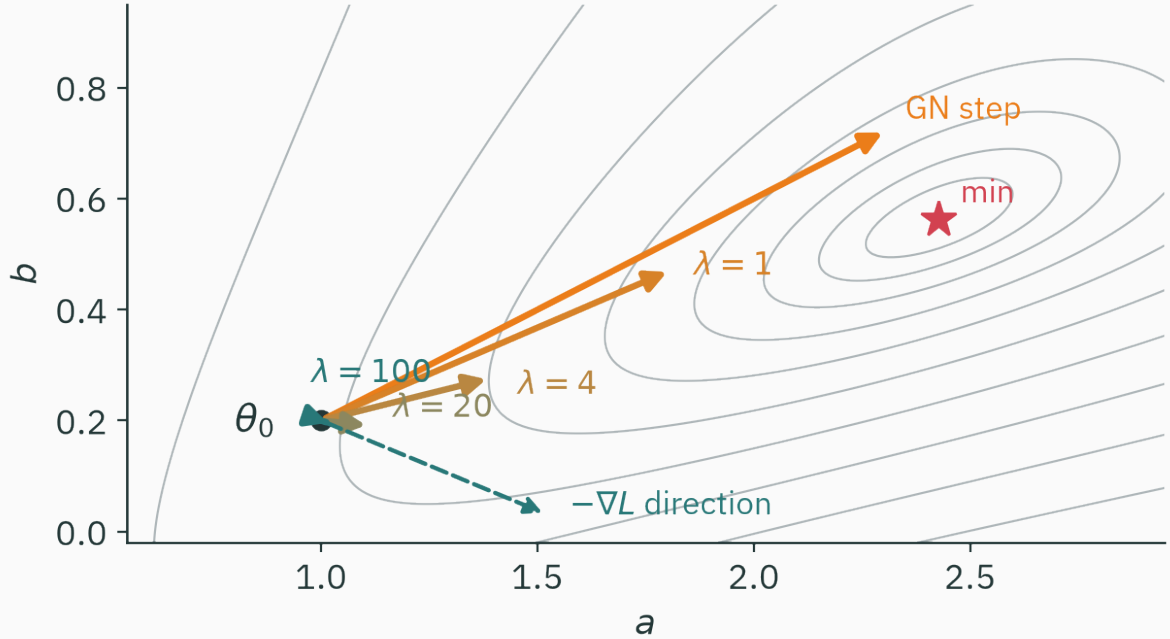
- $\lambda \to 0$: the damping vanishes — **pure Gauss-Newton**, full quadratic confidence.
- λ large: λI dominates $J^\top J$, so $\boldsymbol{\delta} \approx -\frac{1}{\lambda} J^\top \boldsymbol{r} = -\frac{1}{\lambda}\nabla\mathcal{L}$ — a **small gradient-descent step**: humble direction, tiny trust.

One scalar interpolates between “jump to the model’s bottom” and “inch downhill”.

How λ changes the step

V VISUAL

All steps solved from the same θ_0 on the real problem — only λ moves:



as λ climbs, the step **shrinks** and **swings** from the Gauss-Newton jump into the $-\nabla\mathcal{L}$ direction — confidence draining continuously out of the model

Adapt λ using the observed loss

λ isn't tuned by you — it's adjusted **by results**, every iteration:

trial step…	**action**	**meaning**
lowered the loss	accept; $\lambda \downarrow (\div 10)$	model told the truth — trust it more, go more Newton
raised the loss	reject; $\lambda \uparrow (\times 10)$	model lied here — retreat toward small GD steps

A feedback controller with one state variable. Far away it behaves like safe GD; near the fit λ collapses and it **finishes** with Gauss-Newton's doubling digits.

This is Section 5′s trust-region idea compressed into a scalar: λ small = big trust region, λ large = tiny one.

Reproduce `curve_fit` on the sensor data

`scipy.optimize.curve_fit(model, t, y, method='lm')` runs exactly this loop — MINPACK's `lmdif`, Fortran from 1980, still fitting the world's sensors:

```
popt, pcov = curve_fit(model, t, y, p0=(1.0, 0.2))
# popt      = [2.427437, 0.562798]      <- curve_fit (LM)
# our GN k=5: [2.427436, 0.562798]      <- this lecture, 8 lines
```

Same answer to six decimals — on this easy, well-started problem LM's λ just stayed near zero and let Gauss-Newton drive.

- You never supplied a derivative: it builds J by **finite differences** (L10's tool, ideal at $d = 2$).

`curve_fit(method='lm')` combines Gauss-Newton, adaptive damping, and finite-difference Jacobians.

Interactive: beyond gradient descent

I INTERACTIVE

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/optimizers-beyond

Compare Newton and BFGS with gradient descent on 2-D objectives; inspect the quadratic, saddle, and poorly initialized cases, then add damping.

Try the ill-conditioned bowl first: it is this lecture's $\kappa = 50$ picture, live.

Checkpoint: large Levenberg–Marquardt damping

Q QUESTION

In Levenberg–Marquardt, cranking λ up makes the step approach…

A. the pure Gauss-Newton step

B. a tiny step along $-\nabla \mathcal{L}$

C. the exact Newton step (with the true Hessian)

D. a step of fixed length in a random descent direction

Answer: the step approaches scaled gradient descent

A ANSWER

Correct: B — a tiny step along $-\nabla\mathcal{L}$

Why: For large λ, $(J^\top J + \lambda I) \approx \lambda I$, so $\boldsymbol{\delta} \approx -\frac{1}{\lambda} J^\top \boldsymbol{r} = -\frac{1}{\lambda}\nabla\mathcal{L}$. The direction approaches the negative gradient and the length shrinks as $1/\lambda$. LM never uses the true Hessian; even at $\lambda = 0$ it uses the Gauss-Newton approximation $J^\top J$.

Summary & what’s next

Lecture 18 — summary

- **Newton** ⭐: minimize the L9 quadratic model → solve $H\boldsymbol{\delta} = -\nabla f$. A matrix-valued learning rate: step $1/\lambda_i$ per eigen-direction — κ is priced in.
- **On quadratics**: one step to the exact minimum, from anywhere, at any κ.
- **Near any minimum**: quadratic convergence — the error squares, digits double.
- **Cost and failure modes**: d^2 memory, $O(d^3)$ solve; saddles and indefinite Hessians require damping, line search, or trust regions.
- **Least squares** $\mathcal{L} = \frac{1}{2}\|\boldsymbol{r}\|^2$: $\nabla\mathcal{L} = J^\top \boldsymbol{r}$, $H = J^\top J + \sum_i r_i \nabla^2 r_i$.
- **Gauss-Newton**: drop the residual term — $(J^\top J)\boldsymbol{\delta} = -J^\top \boldsymbol{r}$: curvature from first derivatives; five solves fit our sensor.
- **Levenberg–Marquardt**: add λI and adapt λ from the observed loss; this is the method used by `curve_fit`.

Lecture 18 — summary

Read before L19 — MML §7.1.3 (context) · Solomon, *Numerical Algorithms*, Ch. 9 — the reference treatment of nonlinear least squares & Gauss-Newton. **T9** this week: Newton and Gauss-Newton by hand and in NumPy, on this lecture's exact data.

Practice problems

Try on paper; verify against `np.linalg.solve` and `curve_fit`.

1. Run one Newton step on $f(x) = x^2 - 4x + 3$ from $x_0 = 0$. Why is one step enough — and what does GD with $\eta = 0.1$ leave undone after 10 steps?
2. On $f(x, y) = x^2 + xy + y^2$ from $(3, -1)$: solve $H\boldsymbol{\delta} = -\nabla f$ by hand and verify you land at $(0, 0)$.
3. For $f(x) = x - \ln x$, show the Newton update is $x_{k+1} = 2x_k - x_k^2$ and that $e_{k+1} = e_k^2$ exactly. How many steps from $x_0 = 0.5$ to 300 correct digits?
4. Run Newton on the double well $f(x) = x^4/4 - x^2/2$ from $x_0 = 0.3$ for three steps. Where is it converging, and what is the sign of f'' along the way?
5. For the model $\hat{y} = a\sin(bt)$: write r_i and both columns of J, then the Gauss-Newton system at a given (a, b).
6. In LM, show $\boldsymbol{\delta}(\lambda) \to -\frac{1}{\lambda}\nabla\mathcal{L}$ as $\lambda \to \infty$. Which **two** properties of the step does λ therefore control simultaneously?

⭐⭐⭐ Why deep learning still runs first-order

★ OPTIONAL

If Newton is this good, why is every LLM trained with Adam-flavored GD (L17 ⭐⭐⭐)?

per step, d parameters	**GD/Adam**	**Newton**
memory	$O(d)$ — a few copies of $\boldsymbol{\theta}$	$O(d^2)$: at $d = 10^9$, H is $\sim 4 \times 10^9$ GB
arithmetic	$O(d)$ past the gradient	$O(d^3) = 10^{27}$ FLOPs — **years** per step
minibatch noise	averages out (L17)	curvature estimated from noise — precision wasted
saddles (everywhere in high-d)	drifts past	attracted to them

⭐⭐⭐ Why deep learning still runs first-order

★ OPTIONAL

The workarounds are a research industry: **L-BFGS** (build a low-rank H^{-1} sketch from gradient history), Hessian-**vector** products Hv at gradient cost (L11 ⭐⭐⭐), and structured-curvature optimizers (K-FAC, Shampoo, Muon). Every one is a way to **respect** H without ever writing it down — Gauss-Newton's move, philosophically, all over again.

Newton doesn’t step downhill — it teleports to the bottom of its local bowl.

Next: **Constrained Optimization: Lagrange Multipliers** — so far the search was free; real problems have fences.